OligoTox-Coagulopathy — Sources & Provenance

Every source, the database it came from, and a resolvable link · NIH/NCATS Oligonucleotide Toxicity Open Data Challenge, Phase 2

Generated 2026-09-04 from `data/sources.csv`, which is itself generated by the build. Every row of the dataset carries a `source_id` that resolves to an entry below.

100 sources	**8** databases	**2,685** measurements traced	**100** documents held locally

1. Databases and access routes

Every document was retrieved programmatically from a public repository. No value in this dataset comes from a search-engine summary. The endpoints below are the ones actually used, not a generic citation of the resource.

Database	What it holds	Endpoint used	Sources	Rows
PMC	PubMed Central / Europe PMC	`https://www.ebi.ac.uk/europepmc/webservices/rest//fullTextXML`	38	897
PubMed	NCBI PubMed (abstracts)	`https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi?db=pubmed&id=&rettype=abstract`	12	65
USPTO	USPTO granted patents (full text)	`https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/`	10	1,264
Drugs@FDA	FDA Drugs@FDA review packages	`https://www.accessdata.fda.gov/drugsatfda_docs/nda//Orig1s000.pdf`	10	121
EMA	European Medicines Agency EPARs	`https://www.ema.europa.eu/en/medicines/human/EPAR/`	15	130
DailyMed	NLM DailyMed (US prescribing information, SPL)	`https://dailymed.nlm.nih.gov/dailymed/services/v2/spls/.xml`	10	97
openFDA	openFDA (FAERS adverse events, SPL labels)	`https://api.fda.gov/drug/event.json?search=`	4	101
other	other	`–`	1	10

2. How to check any single number

The chain is three steps and needs nothing from us: take a row's `source_id` from `measurements.csv`; find it in the register below or in `sources.csv`; open the link, or open the copy held at `sources/documents/<document_file>` and search for the row's `verbatim_quote` at its `source_locus`. `scripts/verify_against_sources.py` automates exactly this for every numeric value in the release.

3. Link check

Every URL in this document was requested on 2026-09-04: **179 of 185 resolved**. DOIs are checked WITHOUT following the redirect, because the question is whether the DOI is registered; following it would report a publisher's bot-blocking as a broken link. 6 did not. All of them are Google Patents pages returning 503 to automated clients. The authoritative USPTO PDF link for each of those patents resolved, so no patent is unreachable. In every case the copy held at `sources/documents/` is the durable route, which is why the release ships the documents rather than only citing them.

URL	Status
`https://patents.google.com/patent/US10858658`	503
`https://patents.google.com/patent/US7538211`	503
`https://patents.google.com/patent/US8389488`	503
`https://patents.google.com/patent/US9029337`	503
`https://patents.google.com/patent/US9061044`	503
`https://patents.google.com/patent/US9376680`	503

4. Source register

Grouped by the database the document was retrieved from. **Rows** is the number of measurements in the released dataset that cite this source; **Oligos** the number of compounds it describes.

PMC — PubMed Central / Europe PMC (38 sources)

COG-S025 156 rows · 0 oligos

Seth Chhabra E, Liu M. Evaluation of the interference of fitusiran and antithrombin lowering in plasma on routine coagulation assays. Res Pract Thromb Haemost. 2025;9(6):102989.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC12451317/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/40988732/ **resolves** · DOI: https://doi.org/10.1016/j.rpth.2025.102989 **resolves**

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Local copy: `sources/documents/PMC12451317.txt`

COG-S074 80 rows · 32 oligos

Nagano M, Kubota K, Sakata A, Nakamura R, Yoshitomi T, Wakui K, Yoshimoto K. A neutralizable dimeric anti-thrombin aptamer with potent anticoagulant activity in mice. Molecular Therapy Nucleic Acids 2023;33:659-671.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC10445101/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/37621412/ **resolves** · DOI: https://doi.org/10.1016/j.omtn.2023.07.038 **resolves**

Retrieved via: Main text: PMC JATS XML already on disk. Supplementary: the PMC OA service, the PMC /bin/ path and the Europe PMC supplementaryFiles endpoint ALL failed (404 / 500 / HTML error pages - note
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COG-S038 57 rows · 1 oligos

Walsh M, Bethune C, Smyth A, Tyrwhitt J, Jung SW, Yu RZ, Wang Y, Geary RS, Weitz J, Bhanot S; CS4 Investigators. Phase 2 Study of the Factor XI Antisense Inhibitor IONIS-FXIRx in Patients With ESRD. Kidney Int Rep. 2022;7(2):200-209.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC8820988/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/35155859/ **resolves** · DOI: https://doi.org/10.1016/j.ekir.2021.11.011 **resolves** · ClinicalTrials.gov: https://clinicaltrials.gov/study/NCT02553889 **resolves**

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Local copy: `sources/documents/PMC8820988.xml`

COG-S008 41 rows · 10 oligos

Ali AS et al. New High-Affinity Thrombin Aptamers for Advancing Coagulation Therapy. Cells. 2023;12(18):2230.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC10526462/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/37759453/ **resolves** · DOI: https://doi.org/10.3390/cells12182230 **resolves**

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COG-S073 39 rows · 2 oligos

Zavyalova E, Samoylenkova N, Revishchin A, Turashev A, Gordeychuk I, Golovin A, Kopylov A, Pavlova G. The Evaluation of Pharmacodynamics and Pharmacokinetics of Anti-thrombin DNA Aptamer RA-36. Frontiers in Pharmacology 2017;8:922.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC5735248/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/29311929/ **resolves** · DOI: https://doi.org/10.3389/fphar.2017.00922 **resolves**

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Local copy: `sources/documents/PMC5735248.xml`

COG-S039 34 rows · 0 oligos

Buller HR, Bethune C, Bhanot S, Gailani D, Monia BP, Raskob GE, Segers A, Verhamme P, Weitz JI; FXI-ASO TKA Investigators. Factor XI Antisense Oligonucleotide for Prevention of Venous Thrombosis. N Engl J Med. 2015;372(3):232-240.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC4367537/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/25482425/ **resolves** · DOI: https://doi.org/10.1056/NEJMoa1405760 **resolves** · ClinicalTrials.gov: https://clinicaltrials.gov/study/NCT01713361 **resolves**

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Local copy: `sources/documents/NCBI_PMC4367537.xml`

COG-S004 32 rows · 1 oligos

Gissel M, Orfeo T, Foley JH, Butenas S. Effect of BAX499 aptamer on tissue factor pathway inhibitor function and thrombin generation in models of hemophilia. Thromb Res 2012;130(6):948-955.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC3508133/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/22951415/ **resolves** · DOI: https://doi.org/10.1016/j.thromres.2012.08.299 **resolves**

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Local copy: `sources/documents/PMC3508133.xml`

COG-S041 31 rows · 1 oligos

Lang T, Hodel K, Kubitza E, et al. Pharmacokinetics, pharmacodynamics, and safety of fesomersen in healthy Chinese, Japanese, and Caucasian volunteers. Clin Transl Sci. 2024;17(4):e13784.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC10985948/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/38563414/ **resolves** · DOI: https://doi.org/10.1111/cts.13784 **resolves**

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Local copy: `sources/documents/PMC10985948.xml`

COG-S027 29 rows · 0 oligos

Young G, Kavakli K, Klamroth R, et al. Safety and efficacy of a fitusiran antithrombin-based dose regimen in people with hemophilia A or B: the ATLAS-OLE study. Blood. 2025;145(25):2966-2977.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC12824673/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/40053895/ **resolves** · DOI: https://doi.org/10.1182/blood.2024027008 **resolves**

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Local copy: `sources/documents/PMC12824673.txt`

COG-S028 29 rows · 0 oligos

Pipe SW, Lissitchkov T, et al. Long-term safety and efficacy of fitusiran prophylaxis, and perioperative management, in people with hemophilia A or B. Blood Adv. 2025;9(5).

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC11914172/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/39642315/ **resolves** · DOI: https://doi.org/10.1182/bloodadvances.2024013900 **resolves**

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COG-S042 26 rows · 2 oligos

Ferrone JD, Bhattacharjee G, Revenko AS, Zanardi TA, Warren MS, Derosier FJ, Viney NJ, Pham NC, Kaeser GE, Baker BF, Beech S, Monia BP, MacLeod AR, Wang Y, Crooke RM. IONIS-PKKRx a Novel Antisense Inhibitor of Prekallikrein and Bradykinin Production. Nucleic Acid Ther. 2019;29(2):82-91.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC6461157/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/30817230/ **resolves** · DOI: https://doi.org/10.1089/nat.2018.0754 **resolves**

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Local copy: `sources/documents/PMC6461157.xml`

COG-S010 25 rows · 11 oligos

Yu H, et al. Aptameric hirudins as selective and reversible EXosite-ACTive site (EXACT) inhibitors. Nat Commun. 2024;15:3977.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC11087511/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/38730234/ **resolves** · DOI: https://doi.org/10.1038/s41467-024-48211-6 **resolves**

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Local copy: `sources/documents/PMC11087511.xml`

COG-S031 24 rows · 0 oligos

McCluskey G, Maynadie H, Borgel D, et al. Fitusiran treatment modulates the ratio between alpha- and beta-antithrombin isoforms. HemaSphere. 2026;10(5):e70376.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC13146349/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/42100790/ **resolves** · DOI: https://doi.org/10.1002/hem3.70376 **resolves**

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Local copy: `sources/documents/PMC13146349.txt`

COG-S072 23 rows · 4 oligos

Braendli-Baiocco A, Festag M, Dumong Erichsen K, Persson R, Mihatsch MJ, Fisker N, Funk J, Mohr S, Constien R, Ploix C, Brady K, Berrera M, Altmann B, Lenz B, Albassam M, Schmitt G, Weiser T, Schuler F, Singer T, Tessier Y. From the Cover: The Minipig is a Suitable Non-Rodent Model in the Safety Assessment of Single Stranded Oligonucleotides. Toxicological Sciences 2017;157(1):112-128.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC5414856/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/28123102/ **resolves** · DOI: https://doi.org/10.1093/toxsci/kfx025 **resolves**

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Local copy: `sources/documents/PMC5414856.xml`

COG-S007 21 rows · 6 oligos

Soule EE, Yu H, Olson L, Naqvi I, Kumar S, Krishnaswamy S, Sullenger BA. Generation of an anticoagulant aptamer that targets factor V/Va and disrupts the FVa-membrane interaction in normal and COVID-19 patient samples. Cell Chem Biol. 2022 Feb 17;29(2):215-225.e5.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC8808741/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/35114109/ **resolves** · DOI: https://doi.org/10.1016/j.chembiol.2022.01.009 **resolves**

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COG-S023 19 rows · 2 oligos

Abourahma H, Kempaiah P, Farooqui A, Krupa E, et al. A Comparative Study of Porcine and Ovine Derived Defibrotide: Coagulation Profiles, Molecular Characteristics, and DNA Content. Clinical and Applied Thrombosis/Hemostasis. SAGE Publications. doi:10.1177/10760296261429228

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC13009829/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/41869748/ **resolves** · DOI: https://doi.org/10.1177/10760296261429228 **resolves**

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Local copy: `sources/documents/PMC13009829.xml`

COG-S030 18 rows · 0 oligos

Young G, Sorensen B, Dargaud Y, et al. Targeting of antithrombin in hemophilia A or B with investigational siRNA therapeutic fitusiran (ALN-AT3): results of the phase 1 inhibitor cohort. J Thromb Haemost. 2021;19(6):1436-1446.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC8251589/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/33587824/ **resolves** · DOI: https://doi.org/10.1111/jth.15270 **resolves**

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Local copy: `sources/documents/PMC8251589.txt`

COG-S020 17 rows · 1 oligos

Stolte B, Nonnemacher M, Kizina K, Bolz S, Totzeck A, Thimm A, Wagner B, Deuschl C, Kleinschnitz C, Hagenacker T. Nusinersen treatment in adult patients with spinal muscular atrophy: a safety analysis of laboratory parameters. J Neurol. 2021;268(12):4667-4679.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC8563549/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/33899154/ **resolves** · DOI: https://doi.org/10.1007/s00415-021-10569-8 **resolves**

Retrieved via: PMC Open Access JATS XML, staged locally by the triage step
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Local copy: `sources/documents/PMC8563549.xml`

COG-S003 16 rows · 1 oligos

Ay C, Pabinger I, Kovacevic KD, et al. The VWF binding aptamer rondoraptivon pegol increases platelet counts and VWF/FVIII in type 2B von Willebrand disease. Blood Adv 2022;6(18):5467-5476.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC9631691/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/35772170/ **resolves** · DOI: https://doi.org/10.1182/bloodadvances.2022007805 **resolves** · ClinicalTrials.gov: https://clinicaltrials.gov/study/NCT04677803 **resolves**

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Local copy: `sources/documents/PMC9631691.xml`

COG-S018 16 rows · 1 oligos

Maliglowka M, Dec A, Buldak L, Okopien B. The Effects of Inclisiran on the Subclinical Prothrombotic and Platelet Activation Markers in Patients at High Cardiovascular Risk. J Cardiovasc Dev Dis. 2025;12(9):355.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC12470796/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/41002634/ **resolves** · DOI: https://doi.org/10.3390/jcdd12090355 **resolves**

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Local copy: `sources/documents/PMC12470796.txt`

COG-S019 16 rows · 1 oligos

Zhu X, Li H, Hu C, Wu M, Zhou S, Wang Y, Li W. Safety analysis of laboratory parameters in paediatric patients with spinal muscular atrophy treated with nusinersen. BMC Pediatr. 2024;24:474.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC11270951/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/39054521/ **resolves** · DOI: https://doi.org/10.1186/s12887-024-04955-0 **resolves**

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COG-S059 15 rows · 14 oligos

Ohara M, Nagata T, Hara RI, Yoshida-Tanaka K, Toide N, Takagi K, Sato K, Takenaka T, Nakakariya M, Miyata K, Maeda Y, Toh K, Wada T, Yokota T. DNA/RNA heteroduplex technology with cationic oligopeptide reduces class-related adverse effects of nucleic acid drugs. Mol Ther Nucleic Acids. 2024;35(3):102289.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC11382116/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/39252874/ **resolves** · DOI: https://doi.org/10.1016/j.omtn.2024.102289 **resolves**

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Local copy: `sources/documents/PMC11382116.xml`

COG-S029 14 rows · 0 oligos

Kenet G, Nolan B, Zulfikar B, et al. Fitusiran prophylaxis in people with hemophilia A or B who switched from prior BPA/CFC prophylaxis: the ATLAS-PPX trial. Blood. 2024;143(21):2256-2269.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC11181353/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/38452197/ **resolves** · DOI: https://doi.org/10.1182/blood.2023021864 **resolves**

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Local copy: `sources/documents/PMC11181353.txt`

COG-S001 13 rows · 2 oligos

Nimjee SM, de Lange F, Pitoc GA, Sullenger BA. Rats subject to extracorporeal membrane oxygenation have improved cardiac function following anticoagulation and reversal with factor IXa aptamer-antidote oligonucleotide pair. Cardiol Plus 2025;10(2):87-98.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC12208382/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/40599554/ **resolves** · DOI: https://doi.org/10.1097/CP9.0000000000000122 **resolves**

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COG-S011 13 rows · 1 oligos

Reed CR, Bonadonna D, Otto JC, McDaniel CG, Chabata CV, Kuchibhatla M, Frederiksen J, Layzer JM, Arepally GM, Sullenger BA, Tracy ET. Aptamer-based factor IXa inhibition preserves hemostasis and prevents thrombosis in a piglet model of ECMO. Mol Ther Nucleic Acids. 2022;27:524-534.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC8728519/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/35036063/ **resolves** · DOI: https://doi.org/10.1016/j.omtn.2021.12.011 **resolves**

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COG-S040 13 rows · 1 oligos

Crosby JR, Marzec U, Revenko AS, Zhao C, Gao D, Matafonov A, Gailani D, MacLeod AR, Tucker EI, Gruber A, Hanson SR, Monia BP. Antithrombotic effect of antisense factor XI oligonucleotide treatment in primates. Arterioscler Thromb Vasc Biol. 2013;33(7):1670-1678.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC3717325/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/23559626/ **resolves** · DOI: https://doi.org/10.1161/ATVBAHA.113.301282 **resolves**

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Local copy: `sources/documents/NCBI_PMC3717325.xml`

COG-S012 11 rows · 1 oligos

Kolyadko VN, Layzer JM, Perry K, Sullenger BA, Krishnaswamy S. An RNA aptamer exploits exosite-dependent allostery to achieve specific inhibition of coagulation factor IXa. Proc Natl Acad Sci USA. 2024;121(29):e2401136121.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC11260126/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/38985762/ **resolves** · DOI: https://doi.org/10.1073/pnas.2401136121 **resolves**

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COG-S017

10 rows · 2 oligos

Jongejan YK, Dirven RJ, Schrader Echeverri E, de Jong AJL, Pronk ACM, Kooijman S, Rensen PCN, Dahlman JE, Eikenboom JCJ, van Vlijmen BJM. Silencing of the von Willebrand factor gene in proatherothrombotic APOE*3-Leiden.CETP transgenic mice. Res Pract Thromb Haemost. 2025;9(1):102699.

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COG-S043

10 rows · 1 oligos

Liang J, Nilsson S, Wikstrom J, Cao H, Zheng S, Xu Q, Yan X, Ueckert S, Sun Q, Guo C, Zhang H, Liang Z, Gao S, Gan LM. Inhibition of Factor XI Using RBD4059: A Novel GalNAc-siRNA With Potent and Durable Antithrombotic Effects. JACC Basic Transl Sci. 2025;10(6):786-797.

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COG-S061

10 rows · 3 oligos

Aupy P, Echevarria L, Relizani K, Zarrouki F, Haeberli A, Komisarski M, Tensorer T, Jouvion G, Svinartchouk F, Garcia L, Goyenvalle A. Identifying and Avoiding tcDNA-ASO Sequence-Specific Toxicity for the Development of DMD Exon 51 Skipping Therapy. Mol Ther Nucleic Acids. 2020;19:371-383.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC7063478/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/31881528/ **resolves** · DOI: https://doi.org/10.1016/j.omtn.2019.11.020 **resolves**

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COG-S021

9 rows · 1 oligos

Calcaterra IL, Santoro R, Vitelli N, Cirillo F, D'Errico G, Guerrino C, Cardiero G, Di Taranto MD, Fortunato G, Iannuzzo G, Di Minno MND. Assessment of Platelet Aggregation and Thrombin Generation in Patients with Familial Chylomicronemia Syndrome Treated with Volanesorsen: A Cross-Sectional Study. Biomedicines. 2024;12(9):2017.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC11428464/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/39335531/ **resolves** · DOI: https://doi.org/10.3390/biomedicines12092017 **resolves**

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COG-S060

8 rows · 4 oligos

Relizani K, Echevarria L, Zarrouki F, Gastaldi C, Dambrune C, Aupy P, Haeberli A, Komisarski M, Tensorer T, Larcher T, Svinartchouk F, Vaillend C, Garcia L, Goyenvalle A. Palmitic acid conjugation enhances potency of tricyclo-DNA splice switching oligonucleotides. Nucleic Acids Res. 2022;50(1):17-.

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COG-S022

6 rows · 1 oligos

Demirjian S, Ailawadi G, Polinsky M, Bitran D, Silberman S, Shernan SK, Burnier M, Hamilton M, Squiers E, Erlich S, Rothenstein D, Khan S, Chawla LS. Safety and Tolerability Study of an Intravenously Administered Small Interfering Ribonucleic Acid (siRNA) Post On-Pump Cardiothoracic Surgery in Patients at Risk of Acute Kidney Injury. Kidney Int Rep. 2017;2(5):836-843.

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COG-S002

5 rows · 1 oligos

Moreno A, Pitoc GA, Ganson NJ, et al. Anti-PEG antibodies inhibit the anticoagulant activity of PEGylated aptamers. Cell Chem Biol 2019;26(5):634-644.e3.

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COG-S044

5 rows · 1 oligos

Preclinical safety and bleeding evaluation in swine for a small interfering RNA-lipid nanoparticle that prevents excess fibrinogen synthesis. J Thromb Haemost. 2025. (ABSTRACT ONLY - full text not obtainable)

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC12766900/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/41076269/ **resolves** · DOI: https://doi.org/10.1016/j.jtha.2025.09.012 **resolves**

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COG-S065

2 rows · 2 oligos

Schmidt M, Hagner N, Marco A, Koenig-Merediz SA, Schroff M, Wittig B. Design and Structural Requirements of the Potent and Safe TLR-9 Agonistic Immunomodulator MGN1703. Nucleic Acid Ther. 2015;25(3):130-.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC4440985/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/25826686/ **resolves** · DOI: https://doi.org/10.1089/nat.2015.0533 **resolves**

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COG-S066

2 rows · 1 oligos

Peters S, Wirkert E, Kuespert S, Heydn R, Johannesen S, Friedrich A, Mailaender S, Korte S, Mecklenburg L, Aigner L, Bruun TH, Bogdahn U. Safe and Effective Cynomolgus Monkey GLP-Tox Study with Repetitive Intrathecal Application of a TGFBR2 Targeting LNA-Gapmer Antisense Oligonucleotide as Treatment Candidate for Neurodegenerative Disorders. Pharmaceutics. 2022;14(1):200.

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COG-S071 2 rows · 1 oligos

[Author byline not present in the retrieved file and therefore not asserted.] siRNA-mediated reduction of a circulating protein in swine using lipid nanoparticles. Mol Ther Methods Clin Dev. 2024;101258.

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COG-S057 11 rows · 1 oligos

Sheehan JP, Lan HC. Phosphorothioate oligonucleotides inhibit the intrinsic tenase complex. Blood. 1998 Sep 1;92(5):1617-25.

PubMed record: https://pubmed.ncbi.nlm.nih.gov/9716589/ **resolves**

Retrieved via: NCBI eutils PubMed abstract. Blood full text HTTP 403 through the proxy; figures paywalled. Duplicate copies at pubmed_abstracts_mechanism.txt (record 1) and SHEEHAN1998_Blood_pmid9716589_ab

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COG-S062 10 rows · 2 oligos

Henry SP, Novotny W, Leeds J, Auletta C, Kornbrust DJ. Inhibition of coagulation by a phosphorothioate oligonucleotide. Antisense Nucleic Acid Drug Dev. 1997 Oct;7(5):503-10.

PubMed record: https://pubmed.ncbi.nlm.nih.gov/9361909/ **resolves** · DOI: https://doi.org/10.1089/oli.1.1997.7.503 **resolves**

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COG-S058 9 rows · 0 oligos

Sheehan JP, Phan TM. Phosphorothioate oligonucleotides inhibit the intrinsic tenase complex by an allosteric mechanism. Biochemistry. 2001 Apr 24;40(16):4980-9.

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COG-S005 6 rows · 2 oligos

Chan MY, Cohen MG, Dyke CK, et al. Phase 1b randomized study of antidote-controlled modulation of factor IXa activity in patients with stable coronary artery disease. Circulation 2008;117(22):2865-2874.

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COG-S055 6 rows · 1 oligos

Younis HS, Crosby J, Huh JI, Lee HS, Rime S, Monia B, Henry SP. Antisense inhibition of coagulation factor XI prolongs APTT without increased bleeding risk in cynomolgus monkeys. Blood. 2012 Mar 8;119(10):2401-8.

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COG-S063 6 rows · 1 oligos

Shaw DR, Rustagi PK, Kandimalla ER, Manning AN, Jiang Z, Agrawal S. Effects of synthetic oligonucleotides on human complement and coagulation. Biochem Pharmacol. 1997 Apr 25;53(8):1123-32.

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COG-S067 5 rows · 0 oligos

Henry SP, Bolte H, Auletta C, Kornbrust DJ. Evaluation of the toxicity of ISIS 2302, a phosphorothioate oligonucleotide, in a four-week study in cynomolgus monkeys. Toxicology. 1997 Jun 27;120(2):145-55.

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COG-S064 4 rows · 3 oligos

Agrawal S, Rustagi PK, Shaw DR. Novel enzymatic and immunological responses to oligonucleotides. Toxicol Lett. 1995 Dec;82-83:431-4.

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COG-S056 3 rows · 0 oligos

Yacyshyn BR, Barish C, Goff J, Dalke D, Gaspari M, Yu R, Tami J, Dorr FA, Sewell KL. Dose ranging pharmacokinetic trial of high-dose alicaforsen (intercellular adhesion molecule-1 antisense oligodeoxynucleotide) (ISIS 2302) in active Crohn's disease. Aliment Pharmacol Ther. 2002 Oct;16(10):1761-70.

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COG-S068 3 rows · 1 oligos

Burel SA, Han SR, Lee HS, Norris DA, Lee BS, Machemer T, Park SY, Zhou T, He G, Kim Y, MacLeod AR, Monia BP, Lio S, Kim TW, Henry SP. Preclinical evaluation of the toxicological effects of a novel constrained ethyl modified antisense compound targeting signal transducer and activator of transcription 3 in mice and cynomolgus monkeys. Nucleic Acid Ther. 2013 Jun;23(3):213-27.

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COG-S069 1 rows · 0 oligos

Glover JM, Leeds JM, Mant TG, Amin D, Kisner DL, Zuckerman JE, Geary RS, Levin AA, Shanahan WR Jr. Phase I safety and pharmacokinetic profile of an intercellular adhesion molecule-1 antisense oligodeoxynucleotide (ISIS 2302). J Pharmacol Exp Ther. 1997 Sep;282(3):1173-80.

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COG-S070 1 rows · 1 oligos

Sereni D, Tubiana R, Lascoux C, Katlama C, Taulera O, Bourque A, Cohen A, Dvorchik B, Martin RR, Tournerie C, Gouyette A, Schechter PJ. Pharmacokinetics and tolerability of intravenous trecovirsen (GEM 91), an antisense phosphorothioate oligonucleotide, in HIV-positive subjects. J Clin Pharmacol. 1999 Jan;39(1):47-54.

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COG-S035

702 rows · 14 oligos

Modulation of Factor 11 expression. United States Patent US 10,772,906 B2, Ionis Pharmaceuticals, Inc.

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COG-S037

156 rows · 4 oligos

Methods for modulating factor 12 expression. United States Patent US 9,150,864 B2, Isis Pharmaceuticals, Inc.

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COG-S075

129 rows · 8 oligos

Monia BP, Freier SM, Wancewicz EV, et al. (assignee Isis Pharmaceuticals, Inc.). Modulation of transthyretin expression. United States Patent US 9,061,044 B2, granted 23 June 2015. Example 24, 'Effect of ISIS Antisense Oligonucleotides Targeting Human Transthyretin in Cynomolgus Monkeys', subsection 'Coagulation Tests', Tables 86-88.

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COG-S033

87 rows · 10 oligos

Modulation of prekallikrein (PKK) expression. United States Patent US 9,670,492 B2, assignee Ionis Pharmaceuticals, Inc.; application 14/472,318-series filing 28 August 2014, published/granted 06 June 2017. Examples 8, 17, 19, 20, 21 and 22, Tables 50, 52, 78, 81, 82, 83, 84 and 85; per-position chemical notations at lines 1279, 1311 and 1343.

USPTO PDF: https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/9670492 **resolves** · Patent record: https://patents.google.com/patent/US9670492 **resolves**

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COG-S032

69 rows · 2 oligos

Modulation of factor 7 expression. United States Patent US 9,029,337 B2, assignee Isis Pharmaceuticals, Inc.; application 12/741959 (35 U.S.C. 371 national phase of PCT/US2008/082526), filed 05 November 2008, published/granted 12 May 2015. Examples 11-18, Tables 19, 20, 22, 24, 26, 27, 29, 30 and 31.

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COG-S034

55 rows · 21 oligos

Serpinc1 iRNA compositions and methods of use thereof. United States Patent US 9,376,680 B2, assignee Alnylam Pharmaceuticals, Inc.; application 14/806084 (continuation of Ser. No. 13/837,129, now US 9,127,274), filed 22 July 2015, published/granted 28 June 2016. Examples 7-14, Tables 15, 18 and 19; monomer abbreviation table at lines 1840-1960.

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COG-S036

31 rows · 6 oligos

Compositions and methods for inhibiting gene expression of factor XII. United States Patent US 10,858,658 B2, Arrowhead Pharmaceuticals, Inc.

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COG-S014

18 rows · 6 oligos

Crooke S.T. et al. "Antidotes to antisense compounds." United States Patent 8,389,488 B2 (Isis Pharmaceuticals, Inc.), issued 2013-03-05.

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COG-S015

9 rows · 1 oligos

Archemix Corp. "Aptamer therapeutics useful in the treatment of complement-related disorders." United States Patent 7,538,211 B2, issued 2009-05-26.

USPTO PDF: https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/7538211 **resolves** · Patent record: https://patents.google.com/patent/US7538211 **HTTP 503**

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COG-S013

8 rows · 1 oligos

Kandimalla E.R., Manning A., Agrawal S. et al. "Mixed backbone antisense oligonucleotides containing 2'-5'-ribonucleotide- and 3'-5'-deoxyribonucleotide segments." United States Patent 5,886,165 (Hybridon, Inc.), filed 1996-09-24, published 1999-03-23.

USPTO PDF: https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/5886165 **resolves** · Patent record: https://patents.google.com/patent/US5886165 **resolves**

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Drugs@FDA — FDA Drugs@FDA review packages (10 sources)

COG-S093

43 rows · 1 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 219019, QFITLIA (fitusiran) injection, for subcutaneous use. Applicant: Genzyme Corporation. Reference ID 5560526. Approved 28 March 2025.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=219019 **resolves**

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Local copy: `sources/documents/fda_qfitlia_fitusiran_219019_IntegratedR.txt`

COG-S088

20 rows · 0 oligos

Craig EM. Clinical Review (Medical Review), NDA 203568, Kynamro (mipomersen sodium) injection. U.S. Food and Drug Administration, Center for Drug Evaluation and Research, 2012-2013. Reference ID: 3221077. Contains named section 7.3.5.7 'Coagulation and Platelet Issues'.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=203568 **resolves**

Retrieved via: Pre-staged local text conversion of FDA Drugs@FDA NDA 203568 review PDF (fda_kynamro_mipomersen_203568_MedR.pdf, 3225496 bytes, 269 pp); text read with sed/grep, page numbers verified agains

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Local copy: `sources/documents/fda_kynamro_mipomersen_203568_MedR.txt`

COG-S095 15 rows · 1 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 218614, TRYNGOLZA (olezarsen sodium) injection. Applicant: Ionis Pharmaceuticals Inc. Reference ID 5499859. Approved 19 December 2024.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=218614 **resolves**

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COG-S094 12 rows · 0 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Multi-disciplinary Review and Evaluation, NDA 217779, RYTELO (imetelstat) for injection. Applicant: Geron Corporation. Reference ID 5393603. Approved 6 June 2024.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=217779 **resolves**

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Local copy: `sources/documents/fda_rytelo_imetelstat_217779_MultidisciplineR.txt`

COG-S096 11 rows · 2 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 217388, WAINUA (eplontersen) injection. Applicant: Ionis Pharmaceuticals Inc. Reference ID 5299039. Approved 21 December 2023.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=217388 **resolves**

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Local copy: `sources/documents/fda_wainua_eplontersen_217388_IntegratedR.txt`

COG-S080 7 rows · 0 oligos

Mentari E. Clinical Safety Review (Medical Review), NDA 211172, Tegsedi (inotersen). U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Reference ID: 4330662.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=211172 **resolves**

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Local copy: `sources/documents/fda_tegsedi_inotersen_211172_MedR.txt`

COG-S090 6 rows · 0 oligos

Medical Review, NDA 209531, Spinraza (nusinersen) injection. U.S. Food and Drug Administration, CDER, 2016. Bound volume containing the Clinical Safety Review by Evelyn Mentari MD MS (Reference ID 4027874, includes named section 8.4.6.3 'Coagulation Laboratory Measurements'), the Safety Team Leader memorandum (Reference ID 4028770) and the Clinical Review by Rainer W. Paine MD PhD (Reference ID 4026626).

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=209531 **resolves**

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COG-S097 4 rows · 0 oligos

U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Integrated Review, NDA 215887, QALSODY (tofersen) injection, for intrathecal use. Applicant: Biogen. Reference ID 5163540. Approved 25 April 2023.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=215887 **resolves**

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COG-S089 2 rows · 0 oligos

Clinical Pharmacology and Biopharmaceutics Review(s), NDA 203568, Kynamro (mipomersen sodium). U.S. Food and Drug Administration, CDER, 2012. Reference IDs: 3224357 and 3135183. Contains the mipomersen-warfarin drug-drug interaction study MIPO2900509 with INR pharmacodynamic results (Table 10).

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=203568 **resolves**

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COG-S081 1 rows · 0 oligos

Clinical Pharmacology Review, NDA 211172, Tegsedi (inotersen). U.S. Food and Drug Administration, Center for Drug Evaluation and Research. Reference ID: 4282036.

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=211172 **resolves**

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EMA — European Medicines Agency EPARs (15 sources)

COG-S006 22 rows · 3 oligos

Multi-record PubMed abstract file covering the REG1/REG2 (pegnivacogin RB006 / anivamersen RB007) and BAX499 (ARC19499) programmes: PMIDs 21724623, 17101847, 18284597, 20660806, 18506005, 22500821, 19404534, 21503856, 22658294, 22632032, 24263002.

EMA medicine page: https://www.ema.europa.eu/en/medicines **resolves**

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Local copy: `sources/documents/pubmed_REG1_pegnivacogin_BAX499_abstracts.txt`

COG-S009 22 rows · 6 oligos

Yu H, Pitoc G, Zhang M, et al. An Aptamer-Based EXACT Anticoagulant as a Sustainable, Animal-Free Alternative to Unfractionated Heparin. Adv Sci. 2026;13(4):e09867.

PMC article: https://pmc.ncbi.nlm.nih.gov/articles/PMC12822440/ **resolves** · PubMed record: https://pubmed.ncbi.nlm.nih.gov/41053535/ **resolves** · DOI: https://doi.org/10.1002/advs.202509867 **resolves**

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COG-S082 21 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Dawnzera (donidalorsen). 2025.

EMA medicine page: https://www.ema.europa.eu/en/medicines **resolves**

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COG-S076 17 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Waylivra (volanesorsen), International non-proprietary name: volanesorsen, Procedure No. EMEA/H/C/004538/0000. 28 February 2019.

EMA medicine page: https://www.ema.europa.eu/en/medicines **resolves**

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COG-S079 11 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Tegsedi (inotersen), Procedure No. EMEA/H/C/004782/0000. 2018.

EMA medicine page: https://www.ema.europa.eu/en/medicines **resolves**

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COG-S054 7 rows · 6 oligos

NEGATIVE CONTROL (delivery-platform class) - GalNAc-conjugated siRNA US Prescribing Information: GIVLAARI (givosiran), OXLUMO (lumasiran), LEQVIO (inclisiran), AMVUTTRA (vutrisiran), RIVFLOZA (nedosiran), REDEMPLO (plozasiran).

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COG-S077 7 rows · 0 oligos

Akcea Therapeutics Ireland Ltd. WAYLIVRA (volanesorsen) EU Risk Management Plan, CTD Module 1.8.2, Version 3.0, 24 February 2025 (published as part of the Waylivra EPAR).

EMA medicine page: https://www.ema.europa.eu/en/medicines **resolves**

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COG-S053 4 rows · 4 oligos

NEGATIVE CONTROL (chemistry class) - Phosphorodiamidate morpholino oligomer exon-skipping US Prescribing Information: EXONDYS 51 (eteplirsen), VYONDYS 53 (golodirsen), VILTEPSO (viltolarsen), AMONDYS 45 (casimersen). Sarepta Therapeutics / NS Pharma.

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COG-S083 4 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use (CHMP). Assessment report: Amvuttra (vutrisiran). 2022.

EMA medicine page: https://www.ema.europa.eu/en/medicines **resolves**

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COG-S098 4 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use. Assessment report: Qalsody (tofersen). EMA/276404/2024, Procedure EMEA/H/C/005996/0000, 22 February 2024.

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COG-S092 3 rows · 1 oligos

Committee for Medicinal Products for Human Use (CHMP). Assessment report: Spinraza (nusinersen). Procedure No. EMEA/H/C/004312/0000. EMA/289068/2017, 21 April 2017. European Medicines Agency.

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COG-S099 3 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use. Assessment report: Rytelo (imetelstat). EMA/13310/2025, Procedure EMEA/H/C/006098/0000, 12 December 2024.

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COG-S078
2 rows · 0 oligos

European Medicines Agency. Waylivra (volanesorsen) Product Information: Annex I Summary of Product Characteristics, Annex II, Annex III Labelling and Package Leaflet.

EMA medicine page: https://www.ema.europa.eu/en/medicines **resolves**

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COG-S091
2 rows · 1 oligos

Committee for Medicinal Products for Human Use (CHMP). Assessment report: Kynamro (mipomersen), solution for injection 189 mg. Procedure No. EMEA/H/C/002429/0000. EMA/305826/2013, London, 21 March 2013. European Medicines Agency.

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COG-S100
1 rows · 1 oligos

European Medicines Agency, Committee for Medicinal Products for Human Use. Assessment report: Tryngolza (olezarsen). EMA/277774/2025, Procedure EMEA/H/C/006265/0000, 2025.

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DailyMed — NLM DailyMed (US prescribing information, SPL) (10 sources)

COG-S024
34 rows · 1 oligos

DEFITELIO (defibrotide sodium) injection, for intravenous use. US Prescribing Information. Jazz Pharmaceuticals, Inc. Initial U.S. Approval 2016. SPL version 8, effective 2025-07-01 (retrieved via NLM DailyMed).

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=2c3db989-d7ad-41ed-9ebf-698dcf6c24ec **resolves**

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COG-S026
23 rows · 1 oligos

QFITLIA (fitusiran) injection, for subcutaneous use. US Prescribing Information. Genzyme Corporation. SPL version 6, 2025-09-18.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=6dd2f8ac-6f90-4cbf-b197-97d74964135c **resolves**

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COG-S046
9 rows · 1 oligos

RYTELO (imetelstat) for injection, for intravenous use - US Prescribing Information. Geron Corporation. Medication Guide issued 6/2024.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=b0fab7ca-e578-43c5-9df6-bdaff4182257 **resolves**

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COG-S045
8 rows · 1 oligos

TEGSEDI (inotersen) injection, for subcutaneous use - US Prescribing Information. Akcea Therapeutics, Inc. Initial U.S. Approval 2018; label revised 1/2024.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=8513207e-b55f-417b-9473-af785146a543 **resolves**

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COG-S048
7 rows · 1 oligos

DAWNZERA (donidalorsen) injection, for subcutaneous use - US Prescribing Information. Ionis Pharmaceuticals, Inc.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=3ff501e0-f75f-07da-e063-6294a90a0cb7 **resolves**

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COG-S049
7 rows · 1 oligos

SPINRAZA (nusinersen) injection, for intrathecal use - US Prescribing Information. Biogen.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=dd70cd5f-b0fc-4ba4-a5ea-89a34778bd94 **resolves**

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COG-S047
5 rows · 1 oligos

TRYNGOLZA (olezarsen) injection, for subcutaneous use - US Prescribing Information. Ionis Pharmaceuticals, Inc. Patient Information revised 06/2026.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=0f51aa8e-8475-8cf9-e063-6394a90a6848 **resolves**

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COG-S052
2 rows · 1 oligos

NEGATIVE CONTROL - ONPATTRO (patisiran) injection, lipid complex, for intravenous use - US Prescribing Information. Alnylam Pharmaceuticals.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=e87ec36f-b4b4-49d4-aea4-d4ffb09b0970 **resolves**

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COG-S050
1 rows · 1 oligos

NEGATIVE CONTROL - WAINUA (eplontersen) injection, for subcutaneous use - US Prescribing Information. AstraZeneca / Ionis.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=d7dcb847-71dd-4fff-82d0-d43a465fc096 **resolves**

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COG-S051
1 rows · 1 oligos

NEGATIVE CONTROL - QALSODY (tofersen) injection, for intrathecal use - US Prescribing Information. Biogen.

DailyMed label: https://dailymed.nlm.nih.gov/dailymed/drugInfo.cfm?setid=81356b45-1cb7-4eef-88ea-e44cc18b47c5 **resolves**

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openFDA — openFDA (FAERS adverse events, SPL labels) (4 sources)

COG-S085
57 rows · 13 oligos

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openFDA API: https://open.fda.gov/apis/drug/event/ **resolves**

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Local copy: `sources/documents/faers_coag_signal_table.csv`

COG-S086

32 rows · 0 oligos

Individual FAERS case records (openFDA drug/event) for ten oligonucleotide-drug x coagulation-PT pairs, up to 5 records per pair. Generated by /home/user/coag-staging/human/faers/pull_cases.py.

openFDA API: https://open.fda.gov/apis/drug/event/ **resolves**

Retrieved via: openFDA drug/event API via curl, one query per drug x MedDRA PT pair with &limit=5; the exact query URL is stored in each group's exact_query field

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Local copy: `sources/documents/faers_case_records.json`

COG-S087

7 rows · 0 oligos

Individual FAERS case records (openFDA drug/event) for MACUGEN (pegaptanib sodium) x four coagulation MedDRA preferred terms.

openFDA API: https://open.fda.gov/apis/drug/event/ **resolves**

Retrieved via: openFDA drug/event API via curl using patient.drug.medicinalproduct:"MACUGEN"; the exact query URL is stored in each group's exact_query field

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Local copy: `sources/documents/faers_pegaptanib_cases.json`

COG-S084

5 rows · 1 oligos

Astellas Pharma US, Inc. IZERVAY (avacincaptad pegol) intravitreal solution — US Prescribing Information, SPL version 9, effective 2026-02-27. Retrieved as an openFDA drug/label API record (openFDA index last_updated 2026-09-02).

Drugs@FDA: https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&ApplNo=217225 **resolves**

Retrieved via: openFDA drug/label API (api.fda.gov/drug/label.json), single-record JSON response saved to disk

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Local copy: `sources/documents/label_izervay_avacincaptad.json`

other — other (1 sources)

COG-S016

10 rows · 1 oligos

Teva Pharmaceutical Industries / OncoGenex Pharmaceuticals. "Custirsen treatment with reduced toxicity." United States Patent Application Publication US 2014/0275214 A1, published 2014-09-18.

No resolvable public identifier; identified as: US 2014/0275214 A1

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